A-A-53816
31 January 1989

COMMERCIAL ITEM DESCRIPTION

ELECTROCARDIOGRAPH, SINGLE CHANNEL

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a 12-lead, single channel electrocardiograph machine.

Salient characteristics.

The unit shall be a portable, 12-lead, single channel electrocardiograph capable of both automatic and manual lead switching modes.

Applicable Standards. The unit shall meet all applicable requirements of the following:

ANSI/AAMI EC 11	Standard for Diagnostic Electrocardiograph Devices
AAMI ECGC	Standard for ECG Connectors
UL 544	Standard for Safety for Electric Medical and Dental Equipment

Accessories. The unit shall be provided with the following accessories:

QTY	Description
1	Carrying Case
1	Patient Cable
4	Limb electrode straps
1	Set of Reusable electrodes (4 limb and 1 cup)
12	Chest electrodes
2	ECG gel
3	Rolls of chart paper
1	Set of Spare batteries (Each type used)
1	Spare stylus
1	Spare fuse (if required for operation)
1	ECG ruler

AMSC/NA FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Electrical. The unit shall operate from both 120 VAC, 60 Hz and internal battery power. The battery shall be provided with the unit and shall recharge while the unit is connected to an AC power source. The battery shall be capable of supporting operation for at least 1 hour. Battery recharge shall be less than 8 hours from full discharge.

Dimensions. The unit including all accessories and carrying case shall not exceed dimensions of 12 by 10 by 6 inches (30.5 cm by 25.4 cm by 15.2 cm).

Weight. Weight shall not exceed 8 pounds (3.6 Kg).

Frequency. Frequency response shall be 0.05 to 100 Hz, -3dB.

Input impedance. Input impedance shall be at least 20 megaohms at 60 Hz.

Common Mode Rejection Ratio (CMRR). CMRR shall be at least 100 dB measured at 60 Hz.

Defibrillator protection. The unit shall withstand a defibrillator countershock pulse of up to 400 Joules.

Automatic mode. The unit shall perform the following functions without operator assistance when in the automatic mode of operation:

Lead sequencing and annotating
Baseline correction on centering
Gain selection
Calibration signal recording

Lead selection. The electrocardiograph shall have both automatic and manual lead switching. Lead lengths shall be adjustable from 2 to 8 seconds for recording in the automatic mode.

Gain control. Sensitivity of the ECG shall be adjustable in fixed steps of at least 5, 10 and 20 mm/mV.

Amplitude calibration. The unit shall be provided with a 1 mV calibration signal.

Recorder. The device shall be provided with a single channel thermal stripchart recorder. The recorder shall have a minimum of two chart speeds including 25 and 50 mm/s. Chart speed shall be accurate to within 5 percent. The unit shall utilize chart paper with a 40 to 50 mm useable grid width.

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Markings. The unit shall annotate lead code markings on the paper chart.

Workmanship. The unit shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One electrocardiograph, as specified, constitutes one unit.

Manuals. Two complete sets of Operational and Service Data Manuals shall be provided with each unit. The manuals, handbooks and/or brochures shall contain complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks and/or brochures shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc. pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate with ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Inspection shall be conducted at Inspection level II with an AQL of 2.5 defects per hundred. The unit of product shall be one electrocardiograph as specified.

Tests. Tests shall be conducted to determine compliance with specified requirements. Testing shall include but not be limited to the following:

Using an electronic patient simulator, a series of normal and abnormal ECGs shall be used to determine the unit's ability to record data correctly.

For all fixed gain settings, verify that the calibration pulse correctly reflects the gain setting. Error must be less than 5 percent of the expected or 0.5 mm whichever is greater.

The units ability to correctly annotate all leads shall be verified.

By visual inspection, verify chart speeds are accurate to within 5 percent of setting.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting of rejecting the product.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and or order.

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NOTE: This Commercial Item Description is covered by the following item:

NSN	ITEM IDENTIFICATION
6515-01-278-6335	ELECTROCARDIOGRAPH, Automatic and Manual Lead Switching Modes, 120 Volt 60 Hz, AC or Battery Operated: Records 12 leads 12 in by 10 in by 6 in (30.5 by 254 by 15.2 cm)

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4460

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53816	ELECTROCARDIOGRAPH, SINGLE CHANNEL

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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